

Name: Blanco, Paulyn Anne M.	Elective IV : Assignment #1
Yr & Section: IV - CCSAD	Date: August, 2026

Introduction to DevOps

I. Reading Material:

- Read the article on DevOps from a trusted source (e.g., Atlassian, Red Hat, or Microsoft).
- Watch the introductory video on DevOps from a reputable educational platform (e.g., Coursera, Udemy, or YouTube).
- Review the case study on how a well-known company successfully implemented DevOps to improve their software delivery process.

II. Assignment Tasks:

Defining DevOps

- Write a concise definition of DevOps in your own words.
- Explain the main goals of DevOps and how it differs from traditional software development and IT operations approaches.
- Identify and briefly describe the key principles or practices associated with DevOps (e.g., Continuous Integration, Continuous Delivery, Infrastructure as Code).

I. Introduction to DevOps

DevOps has been defined in different ways depending on how people understand and apply it. Some would say that DevOps is a culture, while others would consider it a philosophy. Even though these two terms have completely different meanings, it still circles on the fact that it has the same goal of improving the way systems are being developed.

From the articles (<https://www.atlassian.com/devops/what-is-devops>) I've read and the YouTube video (<https://www.youtube.com/watch?v=dYe83IGQAk4>) I've watched, I can conclude that the word that best describes DevOps is collaboration. DevOps is about how developers and operations teams work together throughout the software development lifecycle. Instead of working in separate departments with little to no communication, both teams collaborate closely to design, build, test, deploy, and maintain systems more efficiently and effectively.

The main goal of DevOps is to deliver new features and fixes faster while maintaining stability and high availability, so that users can access the service anytime without disruption. It also promotes close collaboration between developers and operations teams, ensuring smoother development cycles and fewer miscommunications between teams. In contrast, the traditional approach separates development and operations entirely. This structure leads to slower delivery, higher risk of errors.

Continuous Integration, Continuous Delivery, and Infrastructure as Code are important practices in DevOps that help teams work more efficiently.

- Continuous Integration means developers regularly combine their code into one project and check if everything still works correctly. (similarly to GitHub).
- Continuous Delivery keeps the application ready so new updates can be released without the need of any delay.
- Infrastructure as a Code allows teams to set up and manage servers by writing code instead of configuring everything manually.

These practices improve the Developers and Operator's collaboration, reduce errors, save time, and help teams deliver better and more reliable software more quickly.